

Section 6D. Principal Component Regression

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Principal Component Regression (PCR)

► Recap:

- The *covariance* of two r.v.'s X_r and X_s is defined as $\sigma_{rs} = \mathbb{E}[(X_r - \mu_r)(X_s - \mu_s)]$ where $\mu_k = \mathbb{E}[X_k]$
- The covariance matrix Σ_X of a random vector $X = [X_1, X_2, \dots, X_p]^\top$ is defined entry-wise as $[\Sigma_X]_{rs} = \sigma_{rs}$; equivalently, we have that

$$\Sigma_X = \mathbb{E}[(X - \mu_X)(X - \mu_X)^\top] \in \mathbb{R}^{p \times p} \text{ with } \mu_X = \mathbb{E}[X] \in \mathbb{R}^p$$

For $\mathbb{E}[X] = \mathbf{0}$, we have that $\Sigma_X = \mathbb{E}[XX^\top]$

Principal Component Regression: Problem Statement

Problem Statement: Generate $M < p$ artificial inputs using linear combinations of the original p inputs

- ▶ Consider a linear model with p input variables: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$
- ▶ Find $M < p$ of artificial input variables, $Z_1, \dots, Z_M$, related to the original variables via linear combinations: $Z_m = \phi_{m1} X_1 + \phi_{m2} X_2 \dots + \phi_{mp} X_p$ for $m = 1, \dots, M$
- ▶ Hence, the artificial and original inputs are related via the following linear transformation:

$$Z = \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_M \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} & \cdots & \phi_{1p} \\ \phi_{21} & \phi_{22} & \cdots & \phi_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{M1} & \phi_{M2} & \cdots & \phi_{Mp} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} = \begin{bmatrix} \Phi_1^T \\ \Phi_2^T \\ \vdots \\ \Phi_M^T \end{bmatrix} X$$

where $\Phi_m^T = [\phi_{m1}, \phi_{m2}, \dots, \phi_{mp}]$ and $X = [X_1, X_2, \dots, X_p]^T$

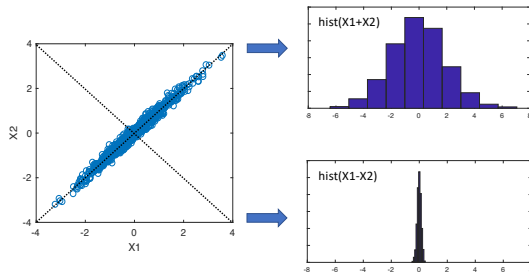
- ▶ With the reduced set of M artificial variables, we would like to perform a linear regression:

$$\hat{Y} = \hat{\theta}_0 + \hat{\theta}_1 Z_1 + \dots + \hat{\theta}_M Z_M$$

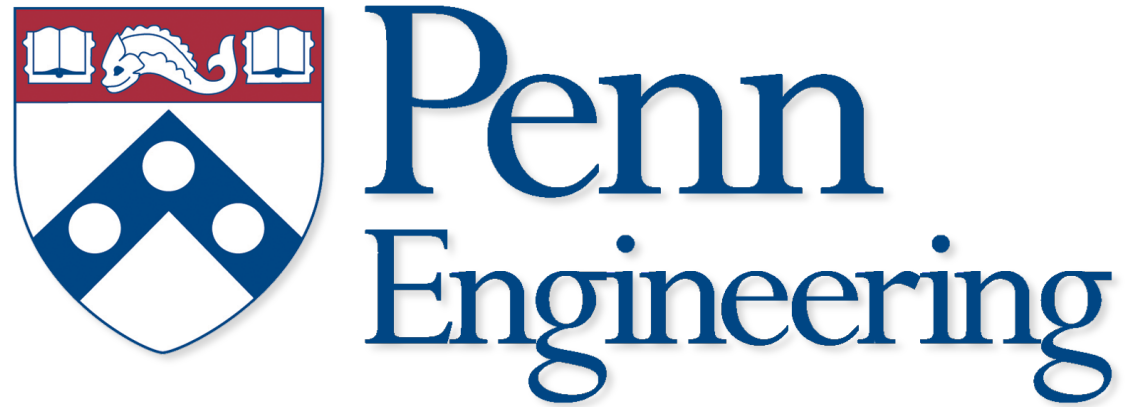
PCR Criterion: Toy Example

Question: How do we choose the transformations Φ_m to achieve a good performance?

- ▶ Consider a dataset with two original inputs, X_1 and X_2 (see figure below)
- ▶ What would be a better choice for a single input variable, $Z_1 = X_1 + X_2$ or $Z_1 = X_1 - X_2$?



PCR Answer: We choose combinations aiming to maximize the variance of the artificial variables. The idea is that, the larger the variance, the more 'information' this variable contains



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